



BobSim

SteadyStateEval Vehicle Characterization

Measured- a_y ramp-steer isolines with robust fits

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Run Notes

- Open-loop ramp-steer at 12.5, 15, 17.5, and 20 m/s
- steerStart = 2.0 s; trim all pre-ramp samples
- Measured a_y is the analysis axis; commanded a_y only sets the ramp
- lmeasured a_y < 1.0 m/s² is excluded; linear band = 1.0 to 4.0 m/s²
- Understeer gradient = mean $d(\text{delta_excess})/d(a_y)$ over the linear band
- Response curves use split Hampel-filtered spline fits; sensitivities trim the outer 20% of lmeasured a_y
- steerExcess = avgSteerAngle - wheelbase * curvature
- Tire/config: MF52, baseline_v3; Backend: OpenModelica

SteadyStateEval Summary 12.5 m/s

Metric	Value	Units
Measured a_y Range	$-16.92 \rightarrow 17.30$	m/s^2
Roadwheel Angle Gradient	0.01034	$\frac{\text{rad}}{\text{m/s}^2}$
	5.81	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.04243	$\frac{\text{rad}}{\text{m/s}^2}$
	23.85	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00200	$\frac{\text{rad}}{\text{m/s}^2}$
	1.12	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00052	$\frac{\text{rad}}{\text{m/s}^2}$
	0.29	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.04243	$\frac{\text{rad}}{\text{m/s}^2}$
	23.85	$\frac{\text{deg}}{g}$
Roll Gradient	1.474	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-18.3	$\text{N} \cdot \text{m}$
Handwheel Torque Max	17.6	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 15.0 m/s

Metric	Value	Units
Measured a_y Range	$-17.34 \rightarrow 18.05$	m/s^2
Roadwheel Angle Gradient	0.00765	$\frac{\text{rad}}{\text{m/s}^2}$
	4.30	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.03142	$\frac{\text{rad}}{\text{m/s}^2}$
	17.66	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00077	$\frac{\text{rad}}{\text{m/s}^2}$
	0.44	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00075	$\frac{\text{rad}}{\text{m/s}^2}$
	0.42	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.03142	$\frac{\text{rad}}{\text{m/s}^2}$
	17.66	$\frac{\text{deg}}{g}$
Roll Gradient	1.479	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-17.6	$\text{N} \cdot \text{m}$
Handwheel Torque Max	16.9	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 17.5 m/s

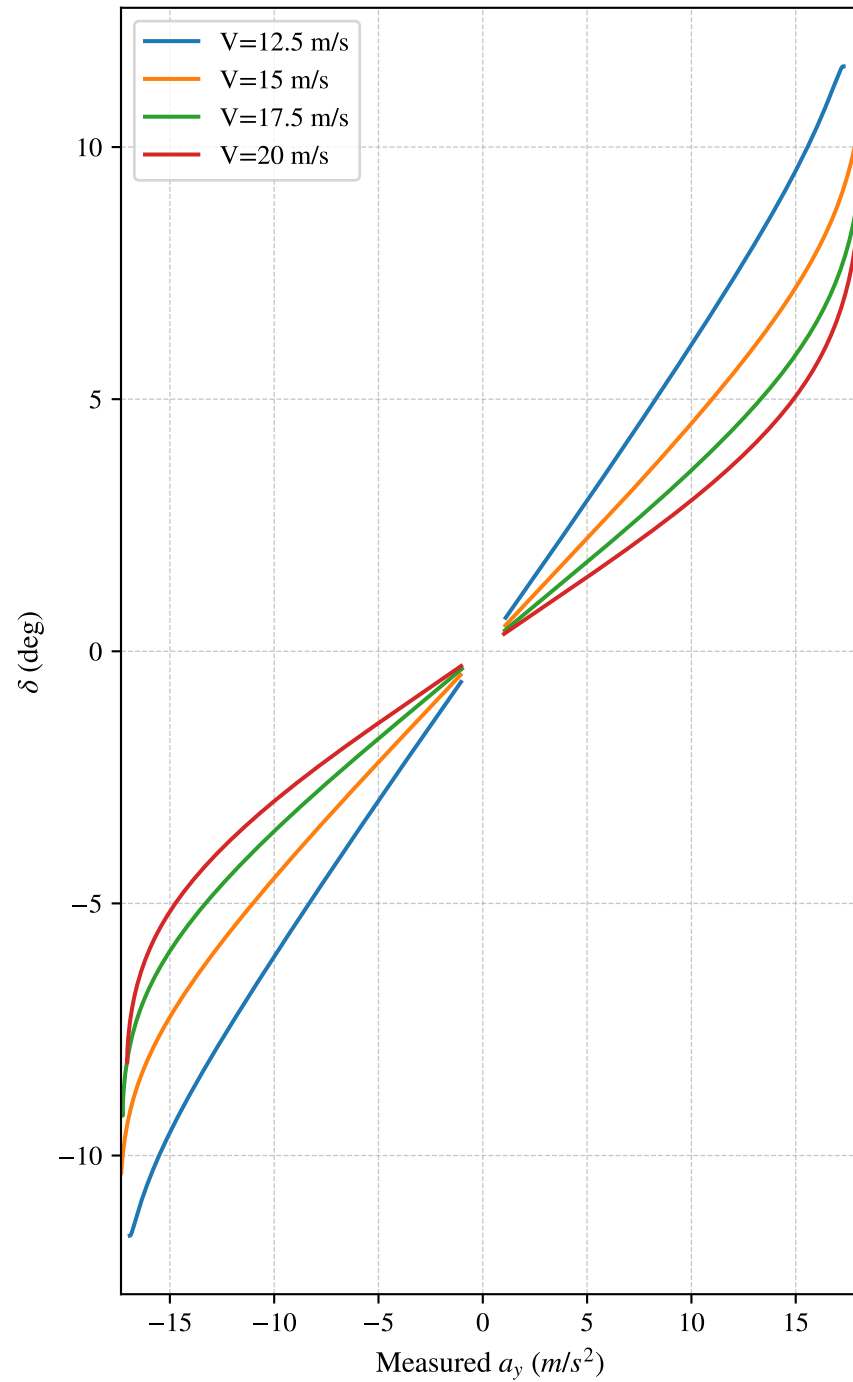
Metric	Value	Units
Measured a_y Range	$-17.26 \rightarrow 18.07$	m/s^2
Roadwheel Angle Gradient	0.00603	$\frac{\text{rad}}{\text{m/s}^2}$
	3.39	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.02474	$\frac{\text{rad}}{\text{m/s}^2}$
	13.91	$\frac{\text{deg}}{g}$
Sideslip Gradient	0.00005	$\frac{\text{rad}}{\text{m/s}^2}$
	0.03	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00087	$\frac{\text{rad}}{\text{m/s}^2}$
	0.49	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02474	$\frac{\text{rad}}{\text{m/s}^2}$
	13.91	$\frac{\text{deg}}{g}$
Roll Gradient	1.488	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-17.1	$\text{N} \cdot \text{m}$
Handwheel Torque Max	16.4	$\text{N} \cdot \text{m}$

SteadyStateEval Summary 20.0 m/s

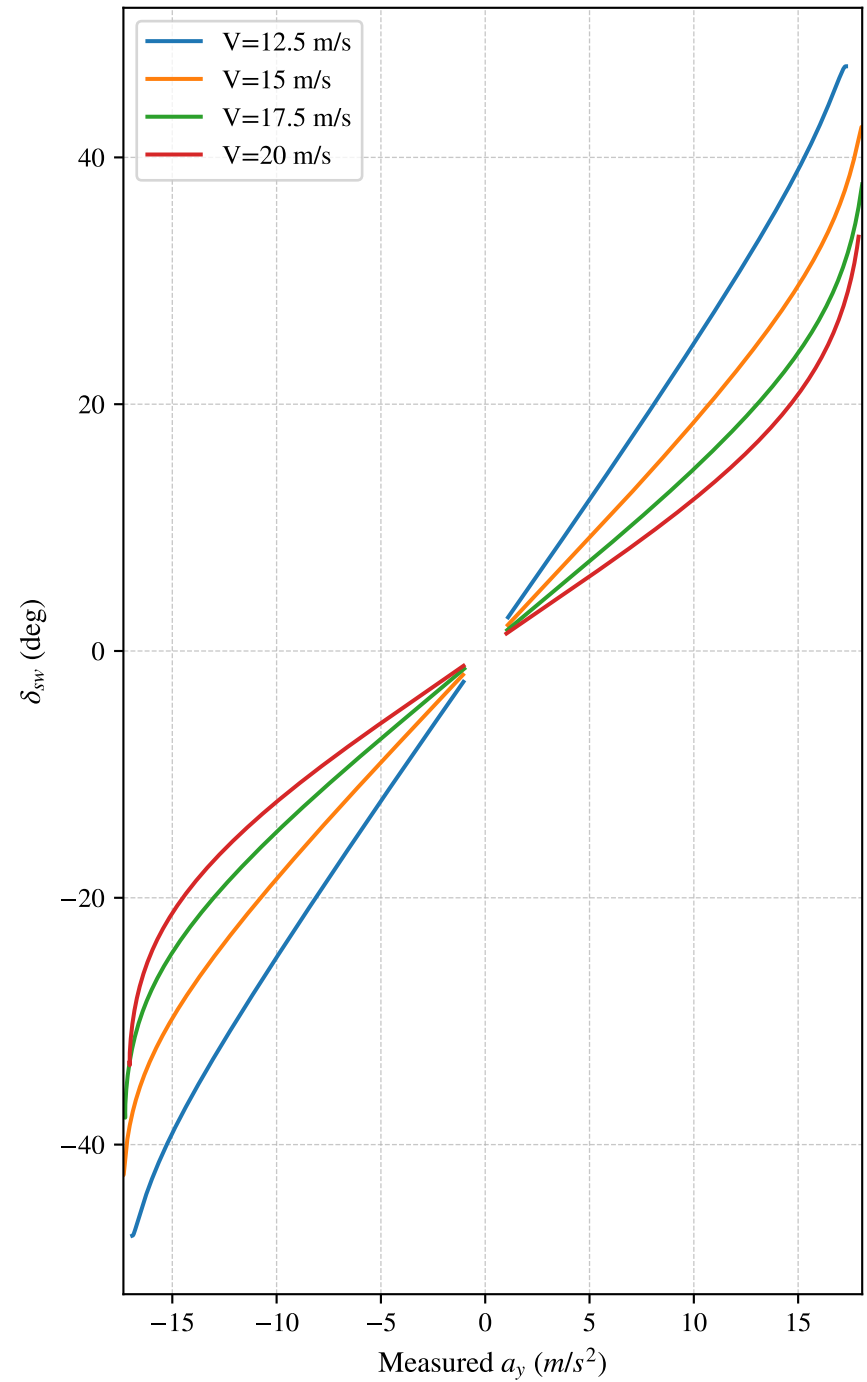
Metric	Value	Units
Measured a_y Range	$-17.05 \rightarrow 17.90$	m/s^2
Roadwheel Angle Gradient	0.00496	$\frac{\text{rad}}{\text{m/s}^2}$
	2.79	$\frac{\text{deg}}{g}$
Handwheel Angle Gradient	0.02037	$\frac{\text{rad}}{\text{m/s}^2}$
	11.45	$\frac{\text{deg}}{g}$
Sideslip Gradient	-0.00041	$\frac{\text{rad}}{\text{m/s}^2}$
	-0.23	$\frac{\text{deg}}{g}$
Understeer Gradient	0.00094	$\frac{\text{rad}}{\text{m/s}^2}$
	0.53	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02037	$\frac{\text{rad}}{\text{m/s}^2}$
	11.45	$\frac{\text{deg}}{g}$
Roll Gradient	1.509	$\frac{\text{deg}}{g}$
Handwheel Torque Min	-16.7	$\text{N} \cdot \text{m}$
Handwheel Torque Max	16.0	$\text{N} \cdot \text{m}$

Steering Angle Isolines vs Measured a_y

Roadwheel Angle δ

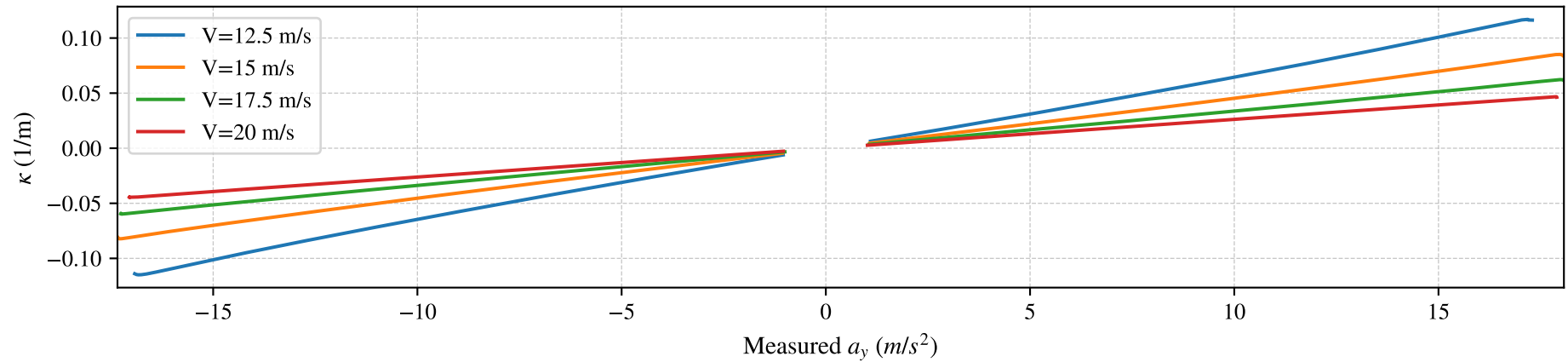


Handwheel Angle δ_{sw}

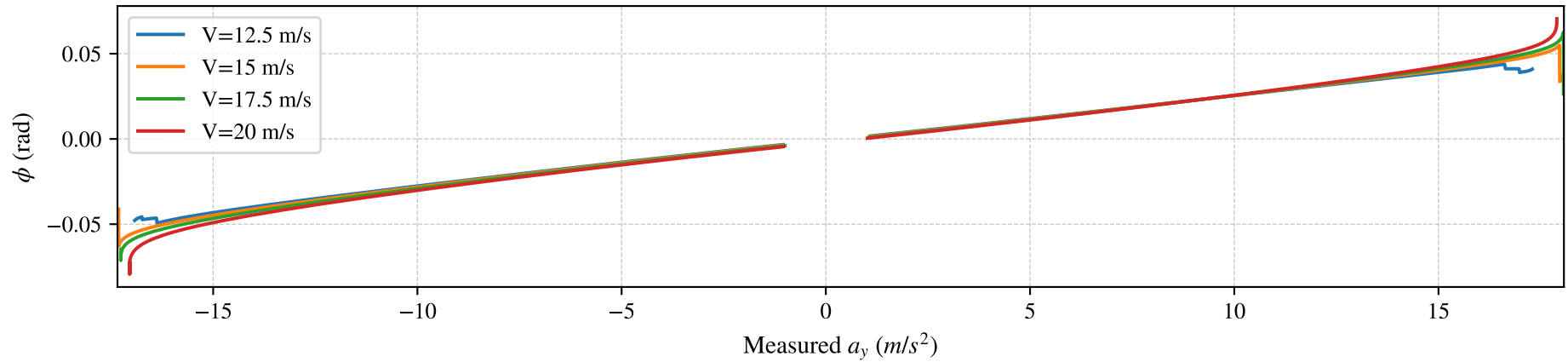


Vehicle State Isolines vs Measured a_y

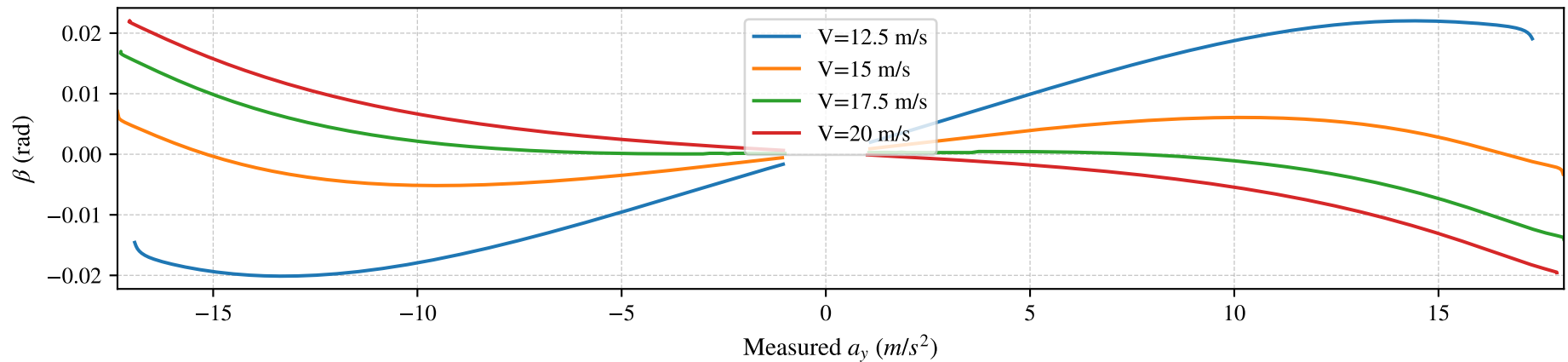
Curvature κ



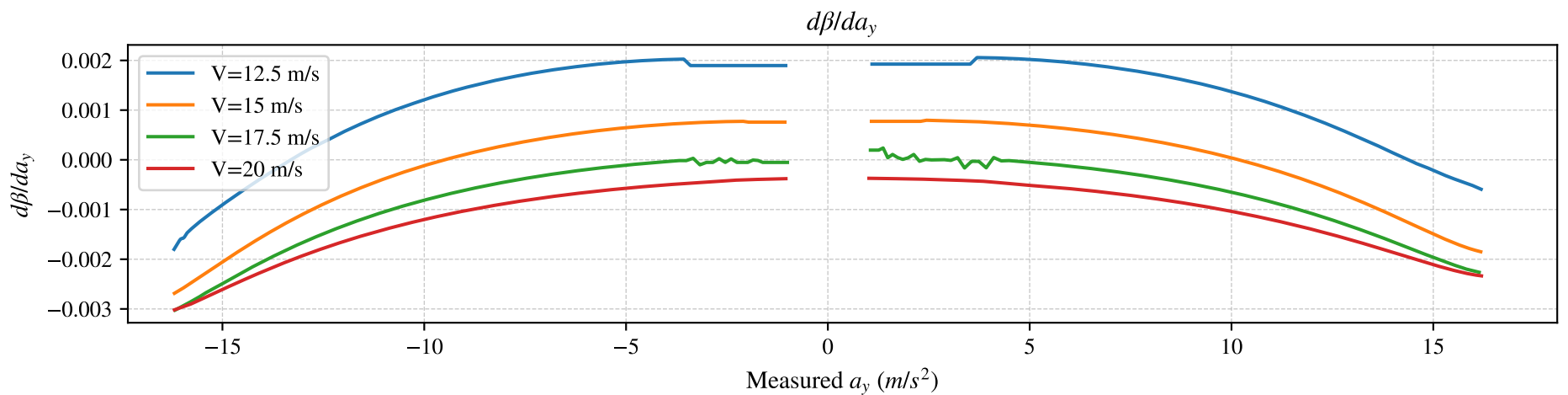
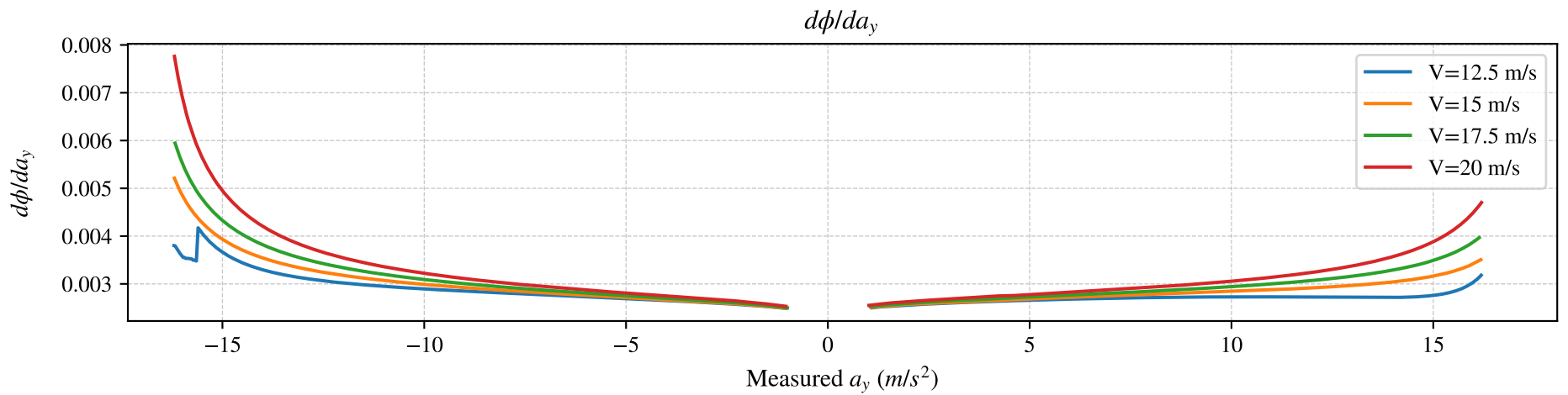
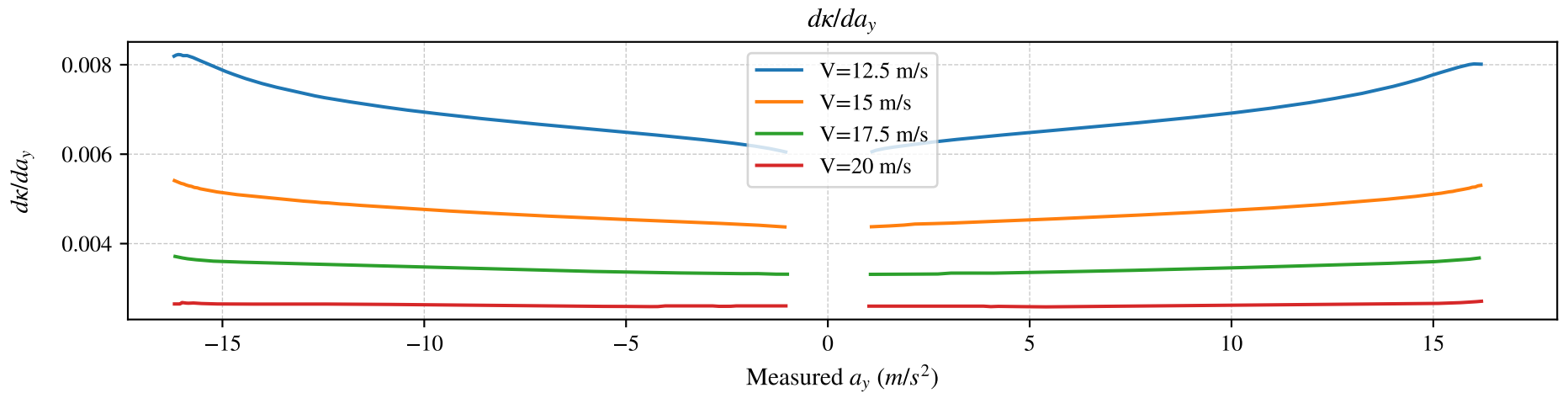
Roll Angle ϕ



Sideslip β

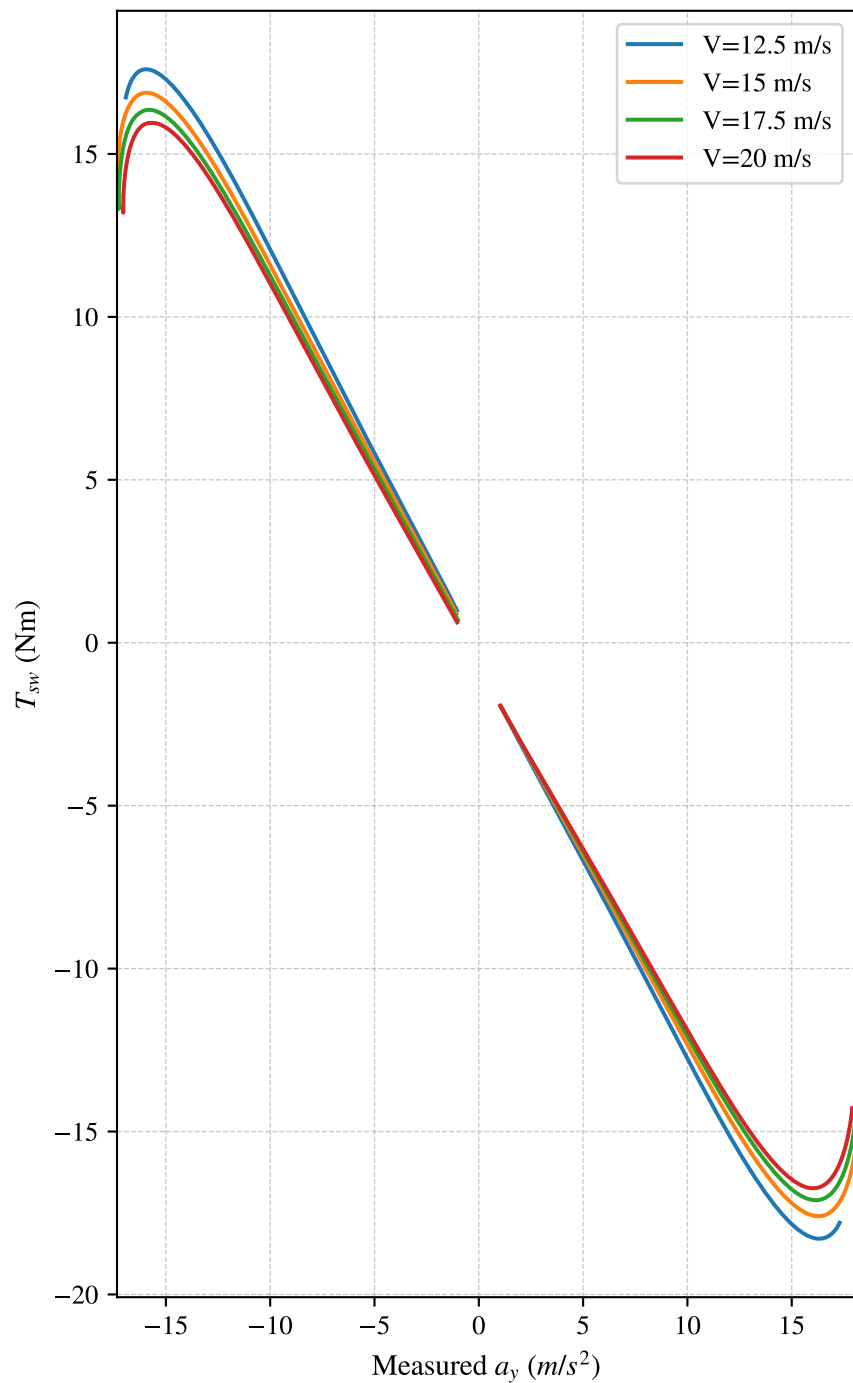


Parametric Response Sensitivities vs Measured a_y

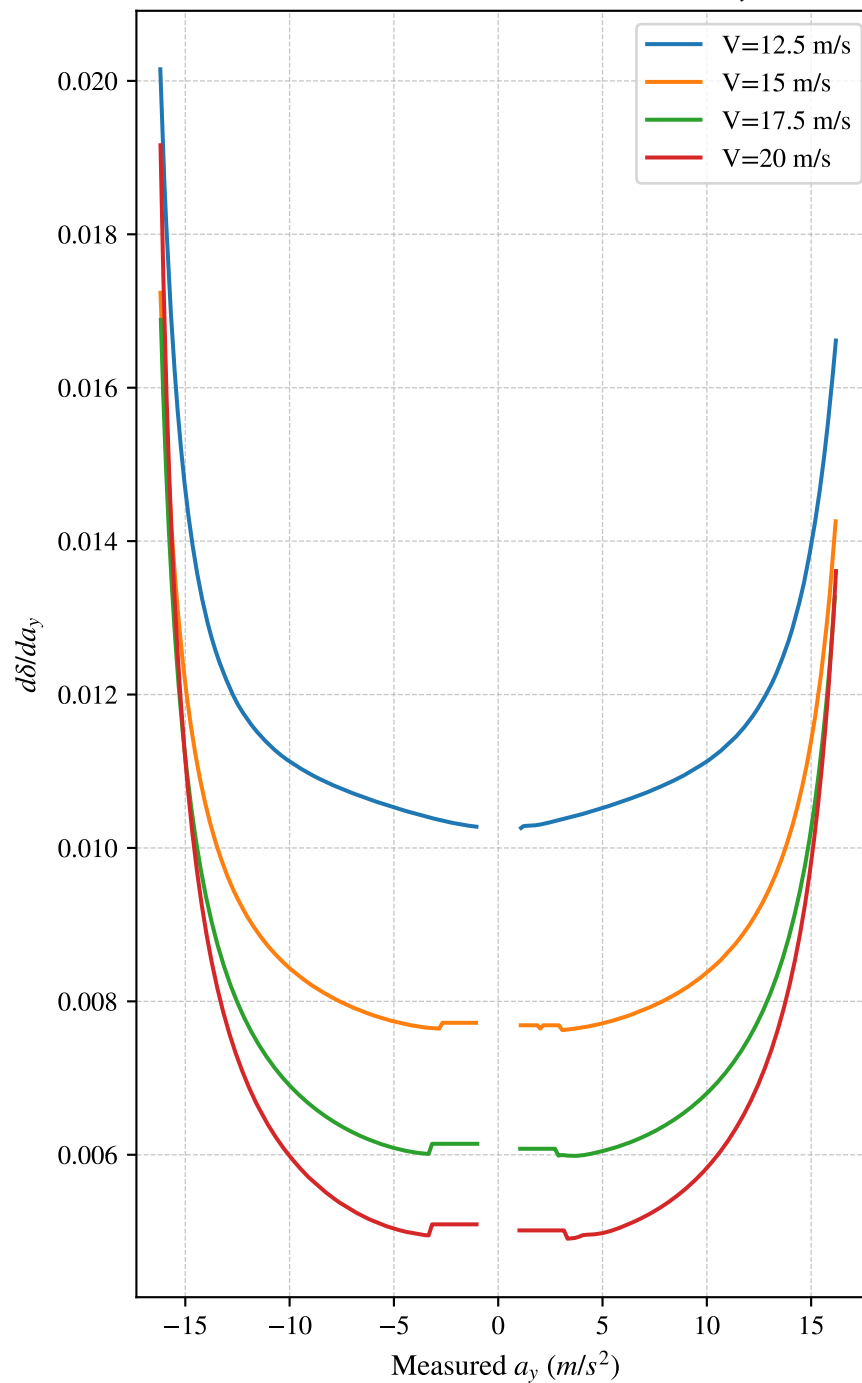


Steering Effort Isolines vs Measured a_y

Steering Torque T_{sw}

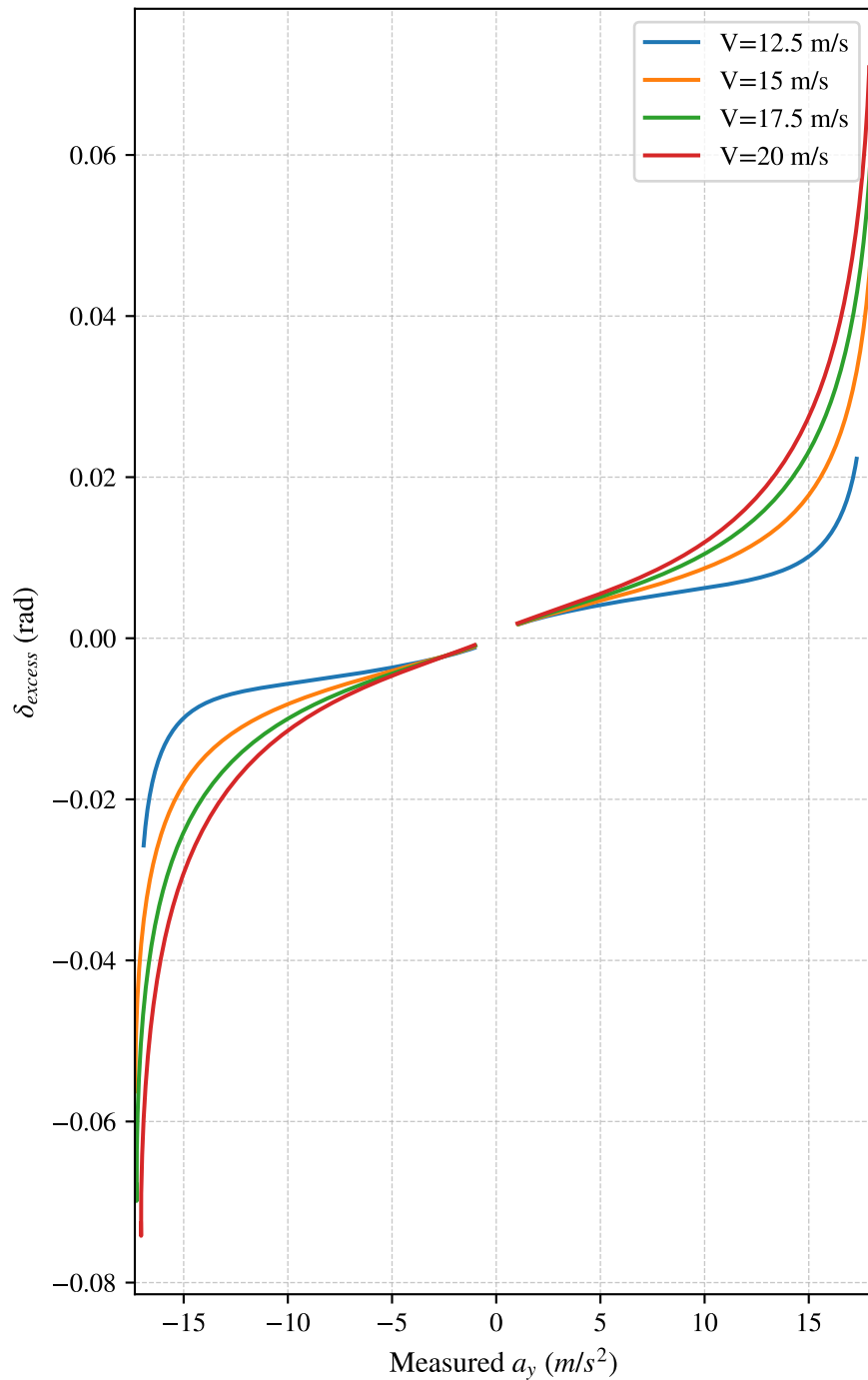


Roadwheel Steering Sensitivity $d\delta/da_y$

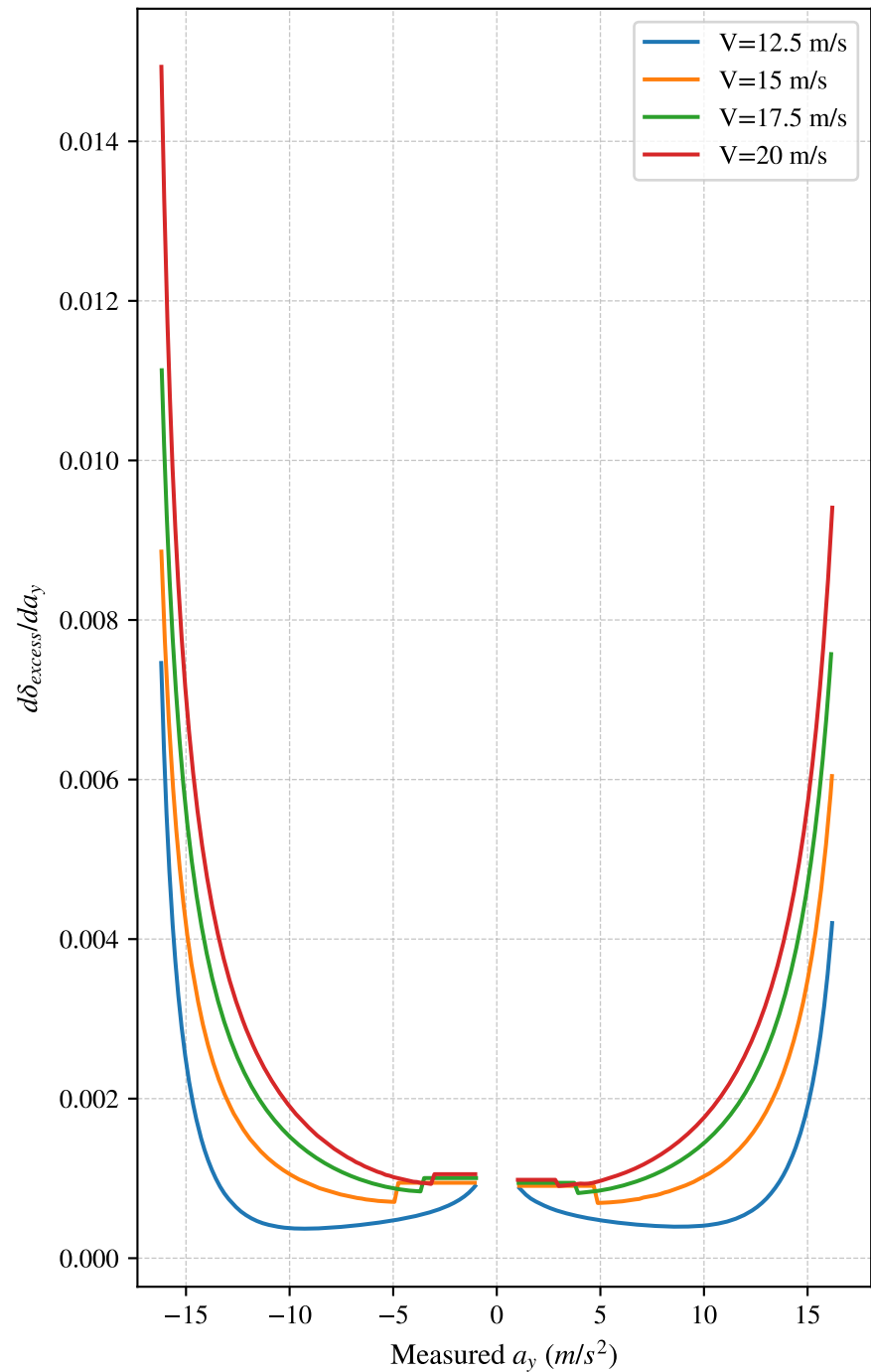


Excess Steer Isolines vs Measured a_y

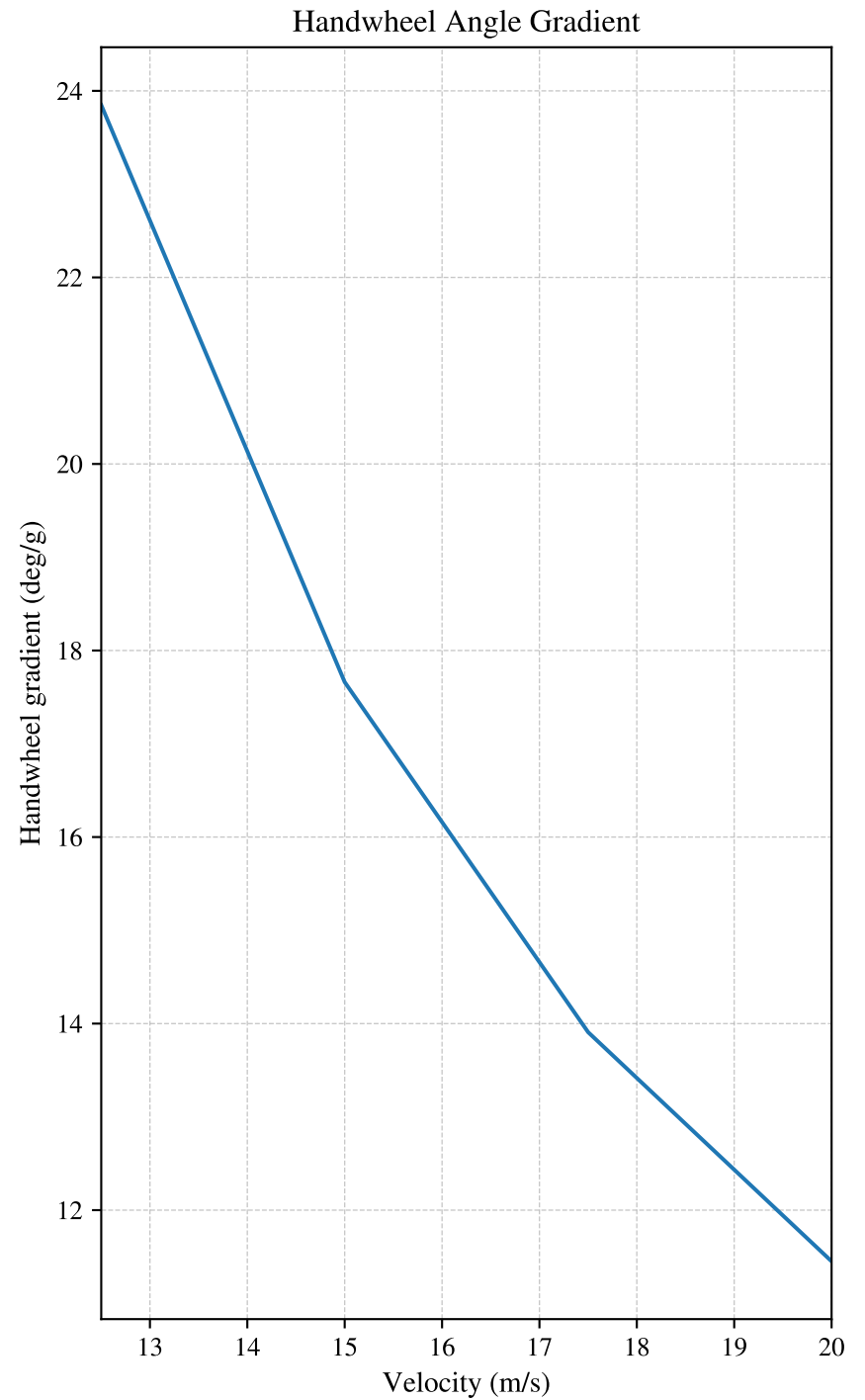
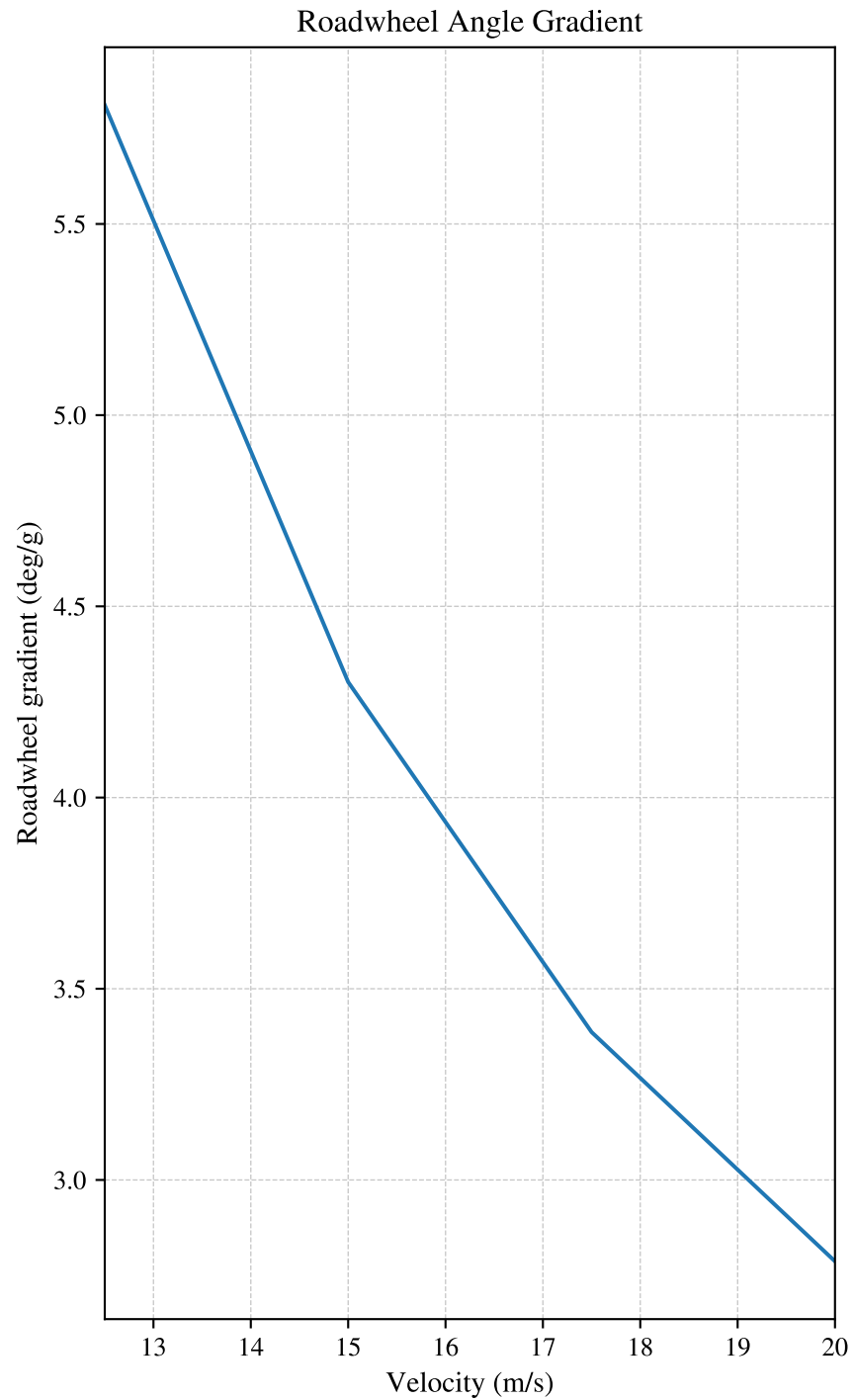
Steer Excess δ_{excess}



Understeer Gradient $d\delta_{excess}/da_y$

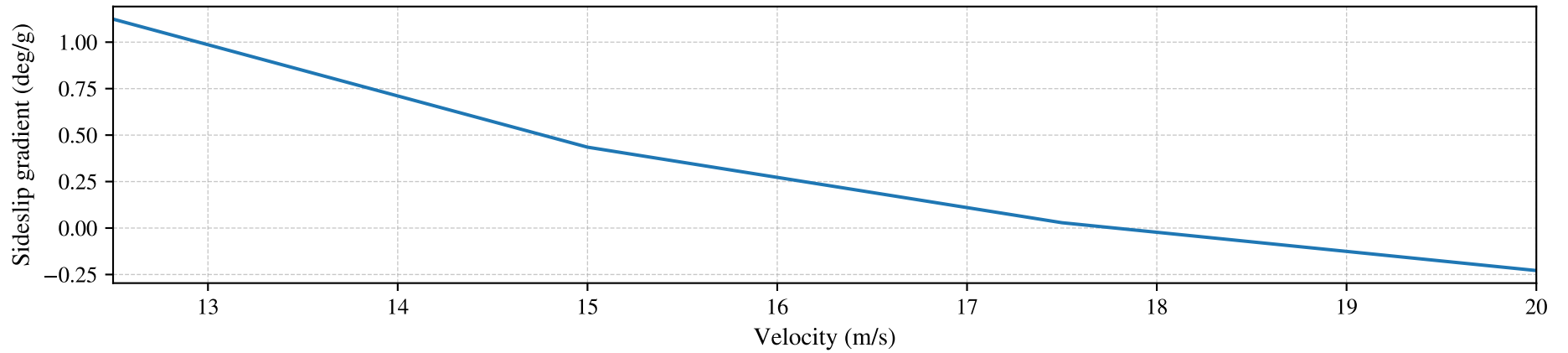


Angle Gradients vs Velocity

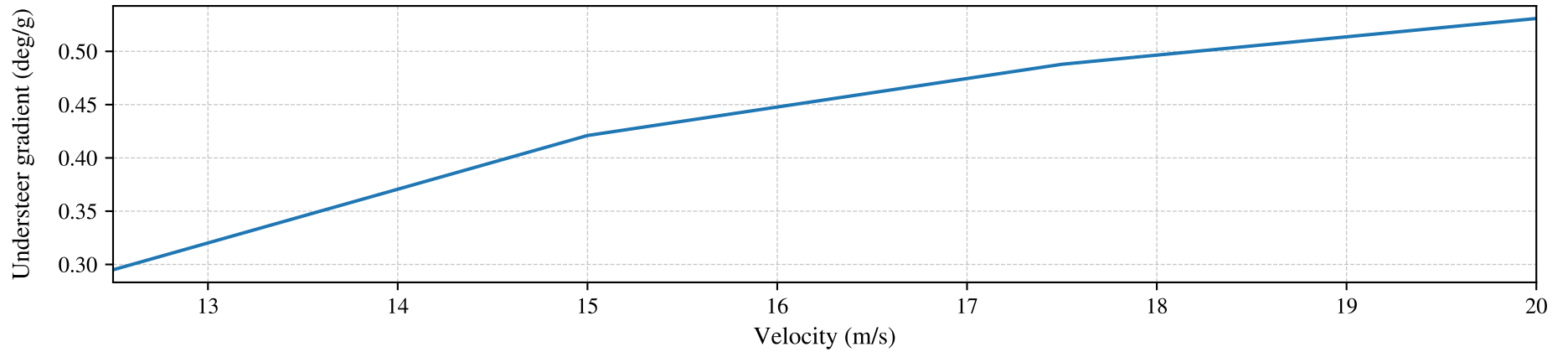


State Gradients vs Velocity

Sideslip Gradient



Understeer Gradient



Roll Gradient

